

simplicial subcomplex S_α^n homeomorphic to S^n at one end of $M(f_\alpha)$, and the homeomorphism $S^n \approx S_\alpha^n$ is homotopic in W_n to the map f_α , hence also to φ_α . Let Z_{n+1} be obtained from Z_n by attaching $D_\alpha^{n+1} \times I$'s via these homotopies between the φ_α 's and the inclusions $S_\alpha^n \hookrightarrow W_n$. Thus Z_{n+1} contains X^{n+1} at one end, and at the other end we have a simplicial complex $Y_{n+1} = Y_n \cup_\alpha C(f_\alpha)$, where $C(f_\alpha)$ is obtained from $M(f_\alpha)$ by attaching a cone on the subcomplex S_α^n . Since $D^{n+1} \times I$ deformation retracts onto $\partial D^{n+1} \times I \cup D^{n+1} \times \{1\}$, we see that Z_{n+1} deformation retracts onto $Z_n \cup Y_{n+1}$, which in turn deformation retracts onto $Y_n \cup Y_{n+1} = Y_{n+1}$ by induction. Likewise, Z_{n+1} deformation retracts onto $X^{n+1} \cup W_n$ which deformation retracts onto $X^{n+1} \cup Z_n$ and hence onto $X^{n+1} \cup X^n = X^{n+1}$ by induction.

Let $Y = \bigcup_n Y_n$ and $Z = \bigcup_n Z_n$. The deformation retractions of Z_n onto X^n give deformation retractions of $X \cup Z_n$ onto X , and the infinite concatenation of the latter deformation retractions is a deformation retraction of Z onto X . Similarly, Z deformation retracts onto Y . $\square$

Exercises

1. What is the minimum number of edges in simplicial complex structures K and L on S^1 such that there is a simplicial map $K \rightarrow L$ of degree n ?
2. Use the Lefschetz fixed point theorem to show that a map $S^n \rightarrow S^n$ has a fixed point unless its degree is equal to the degree of the antipodal map $x \mapsto -x$.
3. Verify that the formula $f(z_1, \dots, z_{2k}) = (\bar{z}_2, -\bar{z}_1, \bar{z}_4, -\bar{z}_3, \dots, \bar{z}_{2k}, -\bar{z}_{2k-1})$ defines a map $f: \mathbb{C}^{2k} \rightarrow \mathbb{C}^{2k}$ inducing a quotient map $\mathbb{CP}^{2k-1} \rightarrow \mathbb{CP}^{2k-1}$ without fixed points.
4. If X is a finite simplicial complex and $f: X \rightarrow X$ is a simplicial homeomorphism, show that the Lefschetz number $\tau(f)$ equals the Euler characteristic of the set of fixed points of f . In particular, $\tau(f)$ is the number of fixed points if the fixed points are isolated. [Hint: Barycentrically subdivide X to make the fixed point set a subcomplex.]
5. Let M be a closed orientable surface embedded in $\mathbb{R}^3$ in such a way that reflection across a plane P defines a homeomorphism $r: M \rightarrow M$ fixing $M \cap P$, a collection of circles. Is it possible to homotope r to have no fixed points?
6. Do an even-genus analog of Example 2C.4 by replacing the central torus by a sphere letting f be a homeomorphism that restricts to the antipodal map on this sphere.
7. Verify that the Lefschetz fixed point theorem holds also when $\tau(f)$ is defined using homology with coefficients in a field F .
8. Let X be homotopy equivalent to a finite simplicial complex and let Y be homotopy equivalent to a finite or countably infinite simplicial complex. Using the simplicial approximation theorem, show that there are at most countably many homotopy classes of maps $X \rightarrow Y$.
9. Show that there are only countably many homotopy types of finite CW complexes.

Chapter 3

Cohomology

Cohomology is an algebraic variant of homology, the result of a simple dualization in the definition. Not surprisingly, the cohomology groups $H^i(X)$ satisfy axioms much like the axioms for homology, except that induced homomorphisms go in the opposite direction as a result of the dualization. The basic distinction between homology and cohomology is thus that cohomology groups are contravariant functors while homology groups are covariant. In terms of intrinsic information, however, there is not a big difference between homology groups and cohomology groups. The homology groups of a space determine its cohomology groups, and the converse holds at least when the homology groups are finitely generated.

What is a little surprising is that contravariance leads to extra structure in cohomology. This first appears in a natural product, called *cup product*, which makes the cohomology groups of a space into a ring. This is an extremely useful piece of additional structure, and much of this chapter is devoted to studying cup products, which are considerably more subtle than the additive structure of cohomology.

How does contravariance lead to a product in cohomology that is not present in homology? Actually there is a natural product in homology, but it takes the somewhat different form of a map $H_i(X) \times H_j(Y) \rightarrow H_{i+j}(X \times Y)$ called the *cross product*. If both X and Y are CW complexes, this cross product in homology is induced from a map of cellular chains sending a pair (e^i, e^j) consisting of a cell of X and a cell of Y to the product cell $e^i \times e^j$ in $X \times Y$. The details of the construction are described in §3.B. Taking $X = Y$, we thus have the first half of a hypothetical product

$$H_i(X) \times H_j(X) \rightarrow H_{i+j}(X \times X) \rightarrow H_{i+j}(X)$$

The difficulty is in defining the second map. The natural thing would be for this to be induced by a map $X \times X \rightarrow X$. The multiplication map in a topological group, or more generally an H-space, is such a map, and the resulting *Pontryagin product* can be quite useful when studying these spaces, as we show in §3.C. But for general X , the only